

image: 'Windows-latest'

created - A temp. windows machine starts up
your code is pulled onto that machine

the build process runs on that machine

install @1

line task that installs
machine before your

Azure

① Pipeline

agents e.g. vms to
run app

white source)

compliance management

security vulnerabilities,

① Pod:

- VmImage: 'Windows-latest'
- Vm is created - A temp. windows machine starts up
- Code is copied - Your code is pulled onto that machine
- Code is built - The build process runs on that machine

② Task: NuGetToolInstaller@1

→ A built-in Azure pipeline task that installs NuGet on the agent machine before your build runs

We need to use microsoft hosted agents e.g. vms to run application & all the tools to run app should be present in agent

Security: Ment Bot tool:- (white source)

open-source also security & compliance management tool.

→ check for any sort of security vulnerabilities, licensing issues.

AKS (Serverless)

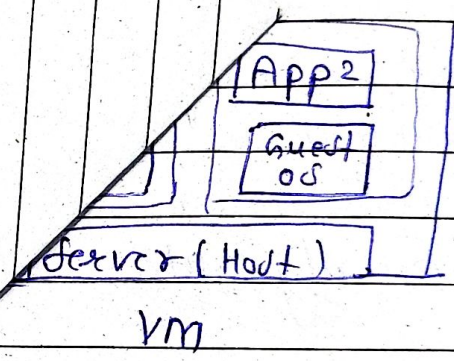
is a fully managed container management
by managed k8s service.
can manage containerized
in the cloud without having
maintain the underlying
infrastructure.

AKS

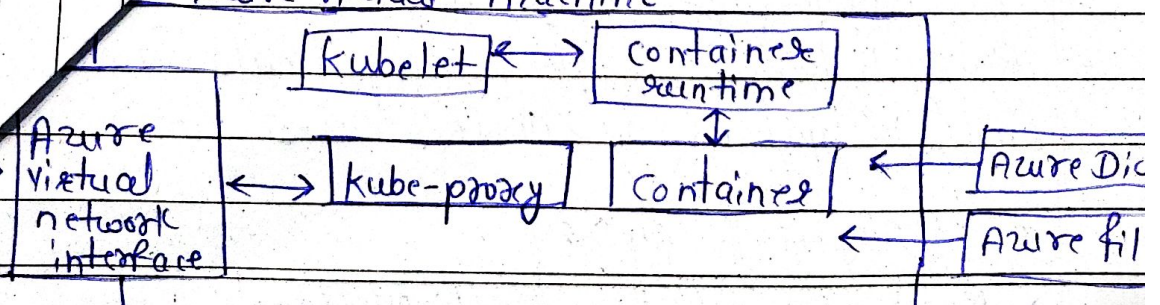
plane (API server, control

plane / apps

security,



charge for worker nodes
Azure virtual machine



Teacher's Signature _____

Teacher's Signature _____

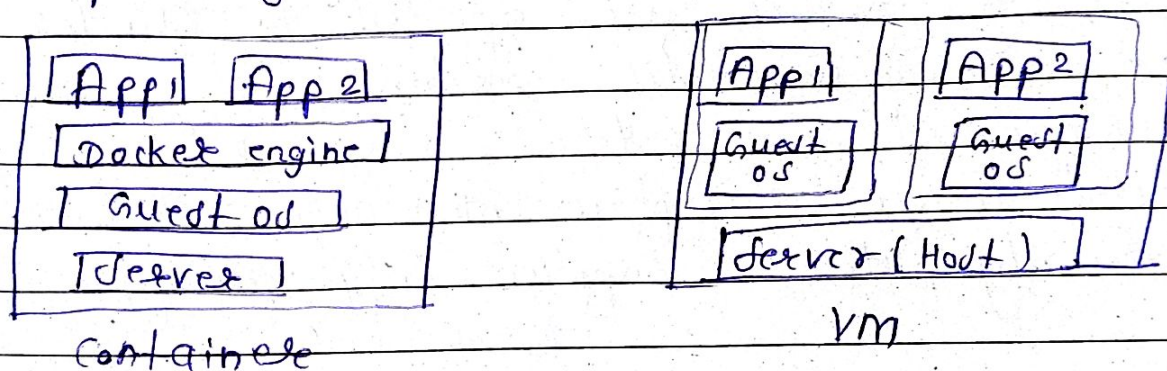
AKS (serverless)

- AKS is a fully managed container management service.
- AKS is fully managed K8s service.
- It lets you run & manage containerized applications in the cloud without having to set up & maintain the underlying K8s infrastructure yourself.

• Azure manages Control plane (API server, control manager, scheduler, etc)

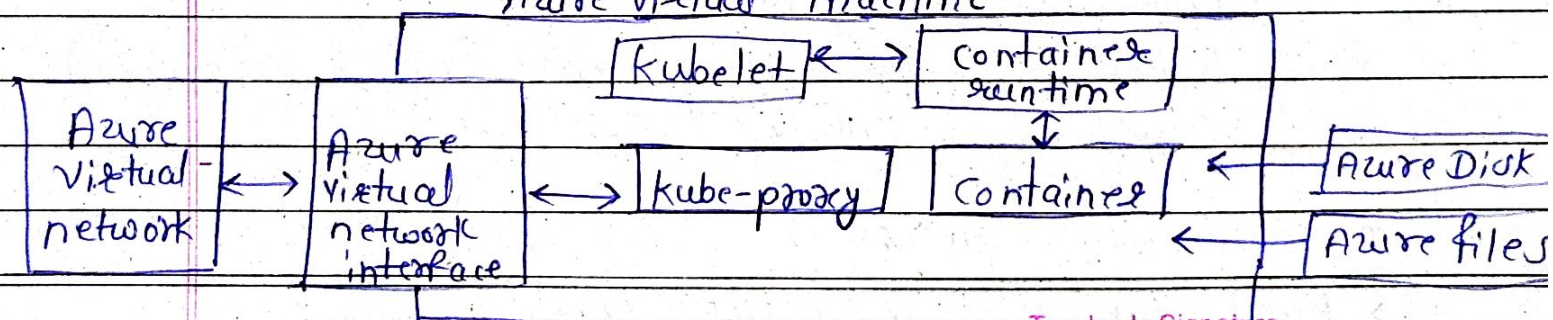
• We only manage workloads/apps

• Azure manages Networking, security, OS patching & node auto-repair.



In container App is independent of OS

→ We only get charge for worker nodes
Azure virtual machine



* Steps to create Azure K8s cluster:-

- ① search for kubernetes service & create a kubernetes cluster

In basic there are 2 options -

- ① Automatic upgrade - Automatically Upgrades kubernetes version when new version release
 - disabled - manual upgrade (good for learning)
 - Patch - auto minor patched only (good for production)
 - Stable/Rapid - more aggressive upgrades.

- ② Node Security channel type - Controls how security patches are applied to cluster nodes.
 - none - no automatic patching
 - unmanaged - os patched only

- * Node Image - replace entire node with fresh patched image
- Security patch - applied only security fixes

Node pool

group of vms (nodes) with same size/config that run your workloads.

Already created -

1. Agentpool (system)

— Runs Kubernetes system components (DND, metrics, etc)
mode - system, Nodes : 2-5

2. Userpool (user)

→ Run your actual applications / pods
mode : user, Nodes : 2-100

We can create user nodes -

① Virtual machine scale set node pool

→ same vm sized

② Add a virtual machine node pool

→ different vm size

We need to select vm size, we can also give count

Then ~~if~~ in monitoring, we need to create log analytic workspace

because it is needed to collect & store logs from cluster.